

Tools of the Trade: Linux and SQL

Activity: Find Files with Linux Commands

Introduction

In this activity, I practiced using basic Linux commands to navigate through directories and find and read files from the Linux file system.

The activity focused on using the Bash shell without a graphical interface. I worked with different directories and files inside the `/home/analyst` directory.

The main commands I practiced were:

- `pwd`
- `ls`
- `cd`
- `cat`
- `head`

These commands are useful for working with files and directories in a Linux environment.

Activity Scenario

In this lab, I worked as a security analyst who needed to locate and analyze information stored in different files.

The files were located under the `/home/analyst` directory.

The tasks involved:

1. Checking the current working directory.
2. Listing the contents of the current directory.
3. Navigating to the reports directory.
4. Finding the users directory.
5. Reading the `Q1_added_users.txt` file.
6. Navigating to the logs directory.
7. Reading the first 10 lines of the `server_logs.txt` file.

This activity helped me practice navigating and reading files through the Linux command line.

Lab Environment

- **Course:** Google Cybersecurity Certificate
 - **Course:** Tools of the Trade: Linux and SQL
 - **Activity:** Find Files with Linux Commands
 - **Operating System:** Linux
 - **Shell:** Bash
 - **User:** `analyst`
 - **Working Directory:** `/home/analyst`
-

Task 1: Get the Current Directory Information

The first task was to find out which directory I was currently working in.

I used the `pwd` command:

```
pwd
```

The output showed:

```
/home/analyst
```

This confirmed that the current working directory was `/home/analyst`.

I then used the `ls` command to see what was inside the directory:

```
ls
```

The directory contained:

```
logs
projects
reports
temp
```

So, the current working directory contained four directories.

The answer to the first question was:

Current working directory: `/home/analyst`

The answer to the second question was:

Number of directories: Four

Task 2: Change Directory and List the Subdirectories

The next task was to move into the `reports` directory.

I used the relative path:

```
cd reports
```

I could also use the full absolute path:

```
cd /home/analyst/reports
```

After entering the `reports` directory, I used:

```
ls
```

The output showed:

```
users
```

This showed that the `reports` directory contained a subdirectory called `users`.

The answer was:

Subdirectory: `users`

Task 3: Locate and Read a File

The next step was to navigate to the `users` directory.

I used:

```
cd /home/analyst/reports/users
```

I then checked the files in the directory using:

```
ls
```

The directory contained the file:

```
Q1_added_users.txt
```

I used the `cat` command to read the contents of the file:

```
cat Q1_added_users.txt
```

The `cat` command displayed the information stored in the file directly in the Bash terminal.

I used the information from the file to answer the questions in the lab.

The employee with the username **aezra** works in:

Human Resources

The employee ID of **mreed**, who works in the Information Technology department, is:

1104

I also learned that the same file could be read using its absolute path:

```
cat /home/analyst/reports/users/Q1_added_users.txt
```

Task 4: Navigate to the Logs Directory and Read a File

The final task was to navigate to the `logs` directory.

I used:

```
cd /home/analyst/logs
```

I then used `ls` to see the file inside the directory:

```
ls
```

The output showed:

```
server_logs.txt
```

I needed to view only the first 10 lines of the file, so I used the `head` command:

```
head server_logs.txt
```

By default, the `head` command displays the first 10 lines of a file.

After reviewing the first 10 lines of the log file, I found that there were:

Three warning messages

This showed how the `head` command can be useful when I only need to quickly check the beginning of a log file instead of displaying the complete file.

Commands Practiced

Command	Purpose
<code>pwd</code>	Display the current working directory
<code>ls</code>	List files and directories
<code>cd</code>	Move between directories
<code>cat</code>	Display the contents of a file
<code>head</code>	Display the beginning of a file

What I Learned

From this activity, I learned how to navigate the Linux file system using the Bash shell.

I practiced finding my current location with `pwd`, viewing files and directories with `ls`, and moving between directories using `cd`.

I also practiced reading complete files with `cat` and checking the first few lines of a file using `head`.

The activity helped me understand how files can be located and reviewed without using a graphical interface.

Cybersecurity Connection

Knowing how to navigate and read files from the Linux command line is important for cybersecurity work.

Security analysts may need to investigate files and logs when looking into a security incident.

For example, log files can contain information about:

- User activity
- Login attempts
- System events
- Warnings
- Errors
- Possible suspicious activity

In this activity, I used the `head` command to review the beginning of a server log and identify warning messages.

I also used `cat` to read information from a user report. These are basic skills that can later be used when working with security logs and investigating incidents.

Practical Skills Demonstrated

- Linux file system navigation
- Bash shell
- Working with directories
- Working with files
- `pwd`
- `ls`
- `cd`
- `cat`
- `head`
- Reading user information
- Reading log files
- Basic command-line investigation

Learning Outcome

This activity gave me practical experience with navigating the Linux file system and reading files through the Bash shell.

I learned how to find my current directory, list files and directories, move between directories, read file contents, and check the beginning of a log file.

These skills are important foundations for cybersecurity because security analysts often need to work with files and logs through the command line.

Demonstration

The following screenshots demonstrate my hands-on work completed during this activity.

The screenshots show the Linux Bash terminal and the commands I used to navigate directories, locate files, read the user report, and examine the server log.

```
analyst@aec6504457b5:~$ pwd
/home/analyst
analyst@aec6504457b5:~$ ls
logs  projects  reports  temp
analyst@aec6504457b5:~$ cd reports
analyst@aec6504457b5:~/reports$ cd /home/analyst/reports
analyst@aec6504457b5:~/reports$ ls
users
analyst@aec6504457b5:~/reports$ cd /home/analyst/reports/users
analyst@aec6504457b5:~/reports/users$ cd users
-bash: cd: users: No such file or directory
analyst@aec6504457b5:~/reports/users$ ls
Q1_added_users.txt  Q1_deleted_users.txt
analyst@aec6504457b5:~/reports/users$
```

```
analyst@aec6504457b5:~/reports/users$ cat Q1_added_users.txt
employee_id  username  department
1001         bmoreno  Marketing
1026         apatel   Human Resources
1041         cgriffin Sales
1104         mreed    Information Technology
1177         aezra    Human Resources
1188         noshiro  Finance
analyst@aec6504457b5:~/reports/users$ cat /home/analyst/reports/users/Q1_added_u
sers.txt
employee_id  username  department
1001         bmoreno  Marketing
1026         apatel   Human Resources
1041         cgriffin Sales
1104         mreed    Information Technology
1177         aezra    Human Resources
1188         noshiro  Finance
analyst@aec6504457b5:~/reports/users$
```

```
analyst@aec6504457b5:~/reports/users$ cd /home/analyst/logs
analyst@aec6504457b5:~/logs$ ls
server_logs.txt
analyst@aec6504457b5:~/logs$ head server_logs.txt
2022-09-28 13:55:55 info    User logged on successfully
2022-09-28 13:56:22 error   The password is incorrect
2022-09-28 13:56:48 warning The file storage is 75% full
2022-09-28 15:55:55 info    User logged on successfully
2022-09-28 15:56:22 error   The username is incorrect
2022-09-28 15:56:48 warning The file storage is 90% full
2022-09-28 16:55:55 info    User navigated to settings page
2022-09-28 16:56:22 error   The password is incorrect
2022-09-28 16:56:48 warning The current user's password expires in 15 days
2022-09-29 13:55:55 info    User logged on successfully
analyst@aec6504457b5:~/logs$
```

The screenshots demonstrate my practical work with:

- `pwd`
- `ls`
- `cd`
- `cat`
- `head`
- Linux directory navigation
- Reading user information
- Reading server logs

Conclusion

This activity helped me become more comfortable with navigating and working with files in a Linux Bash environment.

I practiced the basic commands `pwd`, `ls`, `cd`, `cat`, and `head`.

I also learned how these commands can be useful when looking through user information and server logs, which are common tasks when working with Linux systems and cybersecurity investigations.

Course Information

Google Cybersecurity Certificate

Course: Tools of the Trade: Linux and SQL

Activity: Find Files with Linux Commands

Disclaimer

This document is part of my personal cybersecurity learning portfolio. The activity was completed for educational purposes as part of the Google Cybersecurity Certificate.

The lab was performed in a provided Linux virtual environment, and the screenshots included in this document represent my practical lab work.

